

Registered



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

Continuous Assessment Test (CAT 1) EMT 3202: MECHATRONIC SYSTEM PROGRAMMING II

DATE: December 2025

TIME: 1 HOURS

- a) Explain the function carried out by the RS, RW, EN and data pins i.e. D0 to D7, in a typical 16x2 LCD display (4 marks)
- b) As an embedded engineer who needs to filter input signals, explain and compare the Exponential Moving Average (EMA) filter and the Median Filter, highlighting the pros and cons of each. (6 marks)
- c) Using well labelled diagrams, discuss the operation of optical encoders and magnetic (Hall Effect) encoders. (8 marks)
- d) Using a well labelled diagram discuss the difference between full stepping and half stepping in stepper motor control sequences while stating the main advantage and disadvantage of each method. (6 marks)
- e) You have been asked to design a servo motor control system which takes a reading from a 12 bit Analogue to Digital converter (ADC) and operates a servo motor from -90 degrees to +90 degrees. The minimum control pulse of the servo motor is a PWM signal with an ON time of 0.5ms and a maximum ON time of 1.5 ms. The microcontroller system operates at 16MHz, operating an 8 bit timer to generate the PWM pulse. The timer prescaler values from the data sheet are 1, 8, 64, 256 and 1024. Write a suitable equation that can be used to convert the ADC input to the appropriate PWM count value. (6 marks)

ON $\rightarrow$ 0.5ms